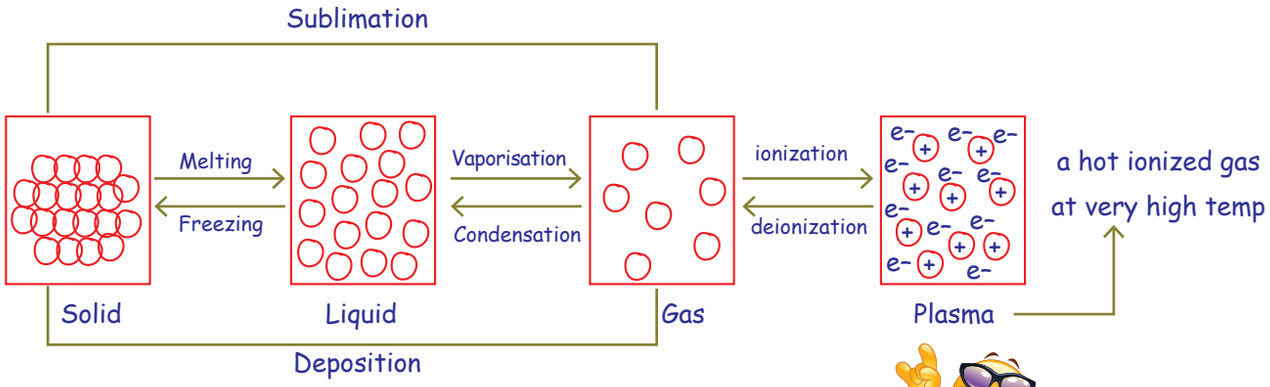
1 Some Basic Concepts of Chemistry

SOLUTION (QR CODE)

4.



Yeh aaya samajh!

7. (B)

- Homogeneous mixtures have a uniform composition, meaning the components are evenly mixed and cannot be distinguished by the naked eye.
- Air (B) is a homogeneous mixture of gases like oxygen, nitrogen, and carbon dioxide. Its composition remains uniform throughout.
- Oil and water, sand and salt, and sand and water are all heterogeneous mixtures, where the components are not uniformly distributed.

yeh agar nahi samajh aaya toh lallu ho yaar

11. (B) In one molecule of $C_{12}H_{22}O_{11}$ → 11-oxygen atoms

In 20 molecules → 11 × 20 = 220 O-atoms

13. For phosphorus, atomic number = 15 = Number of protons

Mass number=31

Mass number = Number of neutrons+number of protons

Number of neutrons = 31 - 15 = 16

For Si, atomic number = 14 = Number of protons = number of electrons (For neutral atom)

Mass number=28

$a^2 + b^2 = (16)^2 + (14)^2 = 452$

yeh bdiya sawaal hai

15. (A) One molecule of CO_2 contains 1 C and 2 O atoms.

For C,

Number of protons = 6

Number of electrons = 6

For O,

Number of protons = 16

Number of electrons = 16

Total number of electrons = 16 + 6 = 22

16. (B) Heavy water is D_2O.

Atomic number of D = 1, for 2-D, no of protons = 2

Atomic number of O = 8, no. of protons = 8

Total number of protons = 2 + 8 = 10

For neutral molecule, number of protons = no. of electrons = 10

Now, mass number of D = 2

So, no of neutrons = Mass number - number of protons = 2 - 1 = 1

For O, no. of neutrons = 16 - 8 = 8

Total no. of neutrons = 2 + 8 = 10

Idhar udhar mat dekho, idhar dhyan lagao

20. According to the given information, reaction can be represented as:

$$\underset{\text{(silver nitrate)}}{AgNO_3} + \underset{\text{(sodium chloride)}}{NaCl} \rightarrow \underset{\text{(silver chloride)}}{AgCl} + \underset{\text{(sodium nitrate)}}{NaNO_3}$$

Let x be the mass of $AgNO_3$

Mass of NaCl = 5.85 g, Mass of AgCl = 14.35 g, Mass of $NaNO_3$ = 8.5 g

Apply law of conservation of mass to the above reaction:

x + 5.85 = 14.35 + 8.5, x = 17 g

reaction toh likho tabhi toh hoga aage

23. According to the given information,

Copper metal + nitric acid → Copper nitrate
2.16 g

$$\text{Copper nitrate} \xrightarrow{\Delta} \underset{2.70\,g}{\text{Copper oxide}}$$

$$\text{Percentage of Cu} = \frac{\text{Mass of Cu Metal}}{\text{Mass of Copper oxide}}$$

$$= \left(\frac{2.16}{2.70}\right) \times 100\% = 80\%$$

Percentage of O = (100- 80)% = 20%

In another experiment,

wt of Cu = 0.92 g

wt of Copper oxide = 1.15 g

Percentage of Cu in copper oxide

$$= \left(\frac{\text{Mass of Cu}}{\text{mass of copper oxide}}\right) \times 100\%$$

$$= \left(\frac{0.92}{1.15}\right) \times 100\% = 80\%$$

Percentage of O = (100 - 80)% = 20%

tedha lag raha hoga

26. (B,C,D) The Law of Multiple Proportions states that when two elements combine to form more than one compound, the different masses of one element that combine with a fixed mass of the other element are in simple whole number ratios. Except option A, all will follow law of multiple proportions.

30. Apply Gay Lussac's law of gaseous volume:

$$H_{2(g)} + Cl_{2(g)} \rightarrow 2HCl_{(g)}$$

1 L 1L 2L

10 L 10L 20 L

So, volume of HCl is 20 L.

33. From the balanced equation:

$$2H_2 + O_2 \rightarrow 2H_2O.$$

2 volumes of H_2 require 1 volume of O_2.

For 10 liters of H_2:

1 Volume of O_2 = $10 \times \frac{1}{2} = 5$ L.

35. Let there be x mL CO in the mixture, hence, there will be: (1000 – x) mL CO_2. The reaction of CO_2 with red hot charcoal may be given as:

$CO_2 + C \rightarrow 2CO$

Total volume of the gas becomes

= x + 2(1000 - x)

x + 2000 - 2x = 1600 L, x = 400 mL

∴ Volume of CO = 400 mL = 0.4 L

Volume of CO_2 = 600 mL = 0.6 L

acha sawaal hai

36. (D) Relative atomic mass = Total nucleons present in the atom (Sum of protons + sum of neutrons)

For C, RAM = 6 + 6 = 12

For H, RAM = 1 + 0 = 1

For N, RAM = 7 + 7 = 14

For Na, RAM = 11 + 12 = 23

Hence, D is the correct option.

40. Weight of the sample (box) = 20 g

weight of one calcium atom (apple) = 40 u

(calcium has 20 p and 20 n i.e. total 40 nucleon)

weight of calcium atom = 40 u = $40 \times 1.66 \times 10^{-24}$ g

$$\text{Number of total calcium atoms} = \frac{\text{wt. of sample}}{\text{wt. of one Ca atom}}$$

$$= \frac{20g}{40 \times 1.66 \times 10^{-24} g} = \frac{1}{2} \times \frac{1}{1.66 \times 10^{-24} g}$$

$$= \frac{1}{2} \times 6.022 \times 10^{23} = \frac{1}{2} N_A$$

[N_A = Avagadro's Number = 6.022×10^{23}]

are muskhil mat samajh lena

42. (a) Total number of protons = Atomic number

For N, number of protons = **7**

No of neutrons = Mass number - no. of protons = 14 - 7 = **7**

No of nucleons = Mass number = **14**

-3 charge on N indicates that it has gained three electrons, so, total number of electrons will become **10**.

No. of valence electrons: 3 gained electrons will be accommodated in valence shell. So they are also considered as valence electrons.

Total valence electrons = 5 + 3 = **8**

(b) NH_4^+

No. of protons = Atomic number of N+ 4 (atomic number of H)

= 7 + 4(1) = **11**

No. of nucleons = Mass number of NH_4^+ = 14 + 4 = **18**

No. of neutrons = Mass number - no. of protons = 18 - 11 = **7**

+ Charge on NH_4 indicates that it has lost one of its electrons, so,

No. of electrons= 11 - 1 = **10**

No. of valence electrons = **8**

(c) CO_3^{2-}

No. of protons = Atomic number of C+ 3 (atomic number of O)

= 6 + 3(8) = **30**

No. of nucleons = Mass number of CO_3^{2-} = 12 + 3(16) = **60**

No. of neutrons = 60 - 30 = **30**

-2 charge indicates that it has gained 2 electrons, so,

No. of electrons = **32**

No. of valence electrons = 4 + 6(3) + 2 = **24**

yeh karlia toh matlab agaya samajh

44. Number of CO_2 molecules $= \frac{50g}{44u}$

$$= \frac{50g}{44 \times 1.66 \times 10^{-24} g} = \frac{50}{44} N_A$$

Total protons = Number of CO_2 molecules × protons in one CO_2 molecule

$$= \frac{50}{44} N_A \times 22 = 25\ N_A$$

Total neutrons = Number of CO_2 molecules × neutrons in one CO_2 molecule

$$= \frac{50}{44} N_A \times 22 = 25\ N_A$$

Total electrons = Number of CO_2 molecules × electrons in one CO_2 molecule

$$= \frac{50}{44} N_A \times 22 = 25\ N_A$$

Here again we see that the weight of the sample was 50 g and total nucleons present

= (25 N_A + 25 N_A) = 50 N_A

thoda hai tedha yeh sawaal, ghabrao mat

45. Step 1: gm हटाओ, N_A लगाओ, Nucleons ले जाओ

Total Nucleons = 150 g N_A

150 N_A

Step 2:

$$\text{Protons} = \frac{\text{Total nucleons}}{2} = \frac{150 N_A}{2} = 75\ N_A$$

$$\text{Neutrons} = \frac{\text{Total nucleons}}{2} = \frac{150 N_A}{2} = 75\ N_A$$

50. $$\text{No. of nucleons} = \frac{\text{Total wt. of sample}}{\text{wt. of one nucleon}}$$

$$= \frac{1g}{1u} = \frac{1g}{1.66 \times 10^{-24} g} = 6.022 \times 10^{23} = N_A$$

gm hatao, N_A lagao → nucleons le jao

56. (B) (A) $\frac{11.2\ Lt}{22.4\ Lt} = \frac{1}{2}$ mole N_2 = 0.5 N_A N_2 molecules = (0.5 N_A) × 2 atoms = N_A atoms

(B) $\frac{24\ g}{(48\ g/mol)} = \frac{1}{2}$ mole O_3 = 0.5 N_A O_3 molecules = (0.5 N_A) × 3 atoms = 1.5 N_A atoms

(C) $\frac{16\ g}{(32\ g/mol)} = \frac{1}{2}$ mole O_2 = 0.5 N_A O_2

molecules = $(0.5\ N_A) \times 2 = N_A$ atoms

(D) $\frac{25\ g}{(100\ g/mol)} = \frac{1}{4}$ mol $CaCO_3 = \frac{1}{4} N_A\ CaCO_3$

molecules $= \left(\frac{N_A}{4}\right) \times 5 = 1.25\ N_A$ atoms

gadbad mat kar dena

85. Let the % abundance of $Cl^{35} = x$

The % abundance of $Cl^{37} = 100 - x$

$$35.5 = \frac{x \times 35 + (100 - x)37}{100}$$

On solving, x = 75%

88. The formula to calculate average molecular weight in terms of H_2 and He is:

$$\text{Avg mol. wt} = \frac{n_{H_2} M_{H_2} + n_{He} M_{He}}{n_{H_2} + n_{He}}$$

$$= \frac{(2 \times 2) + (3 \times 4)}{2 + 3} = \frac{16}{5} = 3.2$$

Formula lagao, sawaal nikalo

90. First calculate moles of each gas present:

Moles of $H_2 = \frac{2}{2} = 1$, Moles of $O_2 = \frac{32}{32} = 1$

Total moles = 1 + 1 = 2

Average molecular weight:

$$M_{avg} = \frac{\sum(\text{moles} \times \text{molecular weight})}{\text{Total moles}}$$

$$= \frac{(1 \times 2) + (1 \times 32)}{2} = \frac{34}{2} = 17 g/mol$$

94. (C) Molar mass of SO_3 = 32 + 48 = 80

Molar mass of CH_4 = 12 + 4 = 16

Ratio of gas 1 wrt 2 = $\frac{P_1 M_1 T_2}{P_2 M_2 T_1}$

$= \frac{M_1}{M_2} = \frac{80}{16} = 5$ [under similar condition]

97. The area of plate = $0.05 \times 10^4\ cm^2$

[1 mt = 10^2 cm, 1 $mt^2 = (10^2)^2\ cm^2 = 10^4\ cm^2$]

Thickness deposited = 0.05 cm

Volume deposited = Area × thickness

$= 0.05 \times 10^4\ cm^2 \times 0.05\ cm = 25\ cm^3$

Mass of Ag deposited = volume × density

$= 25\ cm^3 \times 7.9\ \frac{g}{cm^3} = 197.5$ gm

$$\text{Moles} = \frac{\text{Given mass}}{\text{molar mass}} = \frac{197.5}{108} = 1.83\,\text{mol}$$

Entities (Ag atoms) = moles × N_A

$= 1.83 \times 6.02 \times 10^{23} = 11 \times 10^{23}$

thoda sa hai khatarnaak

99. (D) **Step 1: Write the chemical equation**

$Al + O_2 \rightarrow Al_2O_3$

Step 2: Balance chemical equation

$4Al + 3O_2 \rightarrow 2Al_2O_3$

Step 3: Read the chemical equation

4 moles of Al react with 3 moles of O_2

Step 4: See the data given in the question

1 moles Al react with 3/4 moles of O_2

Mass of O_2 = moles × molar mass

$= \frac{3}{4} \times 32 = 24g$

upar wale steps ki help se kar dia isko

101. Hydrocarbon + $O_2 \rightarrow CO_2 + H_2O$

$$C_aH_b + \left(a + \frac{b}{4}\right)O_2 \rightarrow aCO_2 + \frac{b}{2}H_2O$$

$$C_2H_6 + \left(2 + \frac{6}{4}\right)O_2 \rightarrow 2CO_2 + \frac{6}{2}H_2O$$

$$C_2H_6 + \frac{7}{2}O_2 \rightarrow 2CO_2 + 3H_2O$$

1 mole ethane gives 3 mole of water.

0.1 mole ethane gives 0.3 mole of water.

Molar mass of C_2H_6 = 24 + 6 = 30 g/mol

$$\text{Moles} = \frac{\text{Mass}}{\text{molar mass}} = \frac{3}{30} = 0.1\,\text{mol}$$

No. of molecules = moles × N_A

$= 0.3 \times 6.023 \times 10^{23}$

$= 1.8 \times 10^{23} = 18 \times 10^{22}$, x = 18

general reaction ko yaad karlo,

kaam asaan ho jaega

107. The reaction that occurs is:

$P_4 + 3NaOH + 3H_2O \rightarrow PH_3 + 3NaH_2PO_2$

Moles of phosphorous = $\frac{1.24}{124} = 0.01$ mol.

Since 1 mol of P forms 1 mol of PH_3, 0.01 mol of P will form 0.01 mol of PH_3

$2PH_3 + 3CuSO_4 \rightarrow Cu_3P_2 + 3H_2SO_4$

No. of moles of $CuSO_4$ required to consume 2 mol of phosphine (Q) = 3

So, No. of moles of $CuSO_4$ required to consumed 0.01 mol of phosphine

$(Q) = \left(\frac{3}{2} \times 0.01\right)$ mol

Molar mass of $CuSO_4$ = 159 g/mol

Mass of $CuSO_4$ = $\left(\frac{3}{2} \times 0.01\right) \times 159 = 2.385g$

Jo equation likhoge usko balance karna bhool gye toh tum lallu ho

118. First balance the second reaction:

$3MX_2 + X_2 \rightarrow M_3X_8$

1 mol M_3X_8 = 3 mol MX_2

$\frac{1}{4}$ mol M_3X_8 = $\frac{3}{4}$ mol MX_2

Similarly,

$M_3X_8 + 4N_2CO_3 \rightarrow 8NX + 4CO_2 + M_3O_4$

8 mol NX = 1 mol M_3X_8

1 mol NX = $\frac{1}{8}$ mol M_3X_8

2 mol NX= $\frac{1}{4}$ mol M_3X_8

Moles of MX_2 = $\frac{3}{4}$mol

Now, $M + X_2 = MX_2$

1 mol M = 1 mol MX_2

Molar mass of M = 56 g/mol

Mass of M = $56 \times \frac{3}{4} = 42$ g

time lagega karne mei, tension not

122. (A) First write the balanced chemical reaction according to the given information:

$TiCl_4 + 2Mg \rightarrow Ti + 2MgCl_2$

Molar mass of $TiCl_4$ = 190 g/mol

Moles of $TiCl_4$ = $\frac{\text{Mass}}{\text{molar mass}} = \frac{358}{190}$ mol

Mass of Ti = moles × molar mass

$= \frac{358}{190} \times 48 = 90.44$ g

% yield = $\frac{\text{Actual yield}}{\text{theoretical yield}} \times 100$

$= \frac{32}{90.44} \times 100 = 35.38\%$

dimag ko khula rakho

125. $Pb(NO_3)_2 \rightarrow PbO + 2NO_2 + \frac{1}{2}O_2\uparrow$

ag

$NaNO_3 \rightarrow NaNO_2 + \frac{1}{2}O_2\uparrow$

bg

$\therefore a + b = 5$

Calculate the loss in weight for 5 g mixture as:

$= 5 \times \frac{28}{100} = 1.4$ g

$\therefore$ Residue = 5 – 1.4 = 3.6 g

The residue contains PbO + $NaNO_2$

$\because$ 331g Pb $(NO_3)_2$ gives = 223 g PbO

$\therefore$ ag $Pb(NO_3)_2$ gives = $\frac{223 \times a}{331}$ g PbO

Similarly,

$\because$ 85 g $NaNO_3$ gives = 69 g $NaNO_3$

$\because$ bg $NaNO_3$ gives = $\frac{69 \times b}{85}$ g $NaNO_3$

$\therefore \frac{223 \times a}{331} + \frac{69 \times b}{85} = 3.6$ ...(ii)

From equations (i) and (ii)

a = 3.32 g and b = 1.68 g.

IIT JEE ka hai, isko zarur baithalo dimag ke kone mei, acha sawaal hai

126. Molar mass of C_7H_6O = 106 gm

$A \rightarrow B$

bg ?

1 mol A → 1 mol B

$\frac{5}{92}$ mol A → $\frac{5}{92}$ mol B

$\frac{5}{92} \times 106$ g = 5.76 g

Amount of benzaldehyde = $\frac{5.76 \times 92}{100} = 5.30$ g

136. Molality $\underset{d_{solution}}{\overset{d_{solution}}{\rightleftharpoons}}$ Molarity

6.50 molal KOH

→ 6.5 moles of KOH dissolved in 1000 gm or 1 kg solvent

* moles of solute (KOH) = 6.5 mol

* mass of solvent = 1000 g

* mass of solute = 6.5 × 56 = 364 g

$$\text{Molarity} = \frac{\text{Moles of solute}}{\text{volume of soln in L}}$$

mass of solution = mass of solute + mass of solvent = 364 + 1000 = 1364 g

$$\text{Molality} = \frac{6.5 \times 1000}{\left(\frac{1364}{1.89}\right)} = 9\ M$$

139. mole fraction $\rightleftharpoons$ molality (interconversion)

$$X_{urea} = \frac{1}{31} = \frac{1}{1+30}$$

1 → moles of urea; 30 → moles of water

$$\text{Molality} = \frac{\text{moles of urea}}{\text{wt of solvent in kg}}$$

$$= \frac{1 \times 1000}{540} m = \frac{50}{27} m$$

samajh agaya hoga pakka, bharosa hai

141. Let the mol wt of solute = m

Let the mol wt of solvent = M

$$(\text{mole fraction})_{solute} = \frac{1}{10} = \frac{1}{1+9}$$

$$\text{molality} = \frac{\text{moles of solute}}{\text{wt of solvent in Kg}} = \frac{1 \times 1000}{9M}$$

$$\text{molarity} = \frac{\text{moles of solute}}{\text{vol of solution in lt}}$$

mass of solute + mass of solvent = mass of solution

= (1 × m + 9 × M) = m + 9 M

$$\text{molarity} = \frac{1 \times 1000}{\frac{m+9M}{2}}$$

molality = molarity

$$\frac{1000}{9M} = \frac{2000^{2}}{m+9M},\ m = 9\ M,\ \frac{m}{M} = \frac{9}{1}$$

mushkil toh hai but itna bhi nhi

144. (i) Mass of 1000 mL of $Na_2S_2O_3$ solution = 1.25 × 1000 = 1250 g

Mass of $Na_2S_2O_3$ in 1000 mL of 3 M solution

= 3 × molar mass of $Na_2S_2O_3$ = 3 × 158 = 474 g

Mass percentage of $Na_2S_2O_3$ in solution

$$= \frac{474}{1250} \times 100 = 37.92$$

(ii) No. of moles of $Na_2S_2O_3$ $= \frac{474}{158} = 3$

Mass of water = (1250 – 474) = 776 g

No. of moles of water $= \frac{776}{18} = 43.1$

Mole fraction of $Na_2S_2O_3$

$$= \frac{3}{43.1+3} = \frac{3}{46.1} = 0.065$$

(iii) No. of moles of Na^+ ions

= 2 × No. of moles of $Na_2S_2O_3$ = 2 × 3 = 6

Molality of Na^+ ions

$$= \frac{\text{No. of moles of } Na^+ \text{ ions}}{\text{Mass of water(in kg)}} = \frac{6}{776} \times 1000$$

= 7.73 m

No. of moles of $S_2O_3^{2-}$ ions = No. of mole of $Na_2S_2O_3$ = 3

Molality of $S_2O_3^{2-}$ ions $= \frac{3}{776} \times 1000 = 3.86$ m

badhiya sawaal hai, dekhlo dhyan se

145. 10% w/w aq urea solution

→ 10 gm urea in 100 g solution

wt of urea = 10 g

wt of solution = 100 g

wt of solvent = wt of solution-wt of solute (urea)

wt of solvent = 100 – 10 = 90 g

$$X_{urea} = \frac{n_{urea}}{n_{urea} + n_{H_2O}}$$

$$\text{molality} = \frac{\text{moles of urea}}{\text{wt of } H_2O \text{ in Kg}}$$

$$X_{urea} = \frac{\frac{10}{60}}{\frac{10}{60}+\frac{90}{18}} = \frac{1}{31}$$

$$m = \frac{\left(\frac{10}{60}\right)}{\frac{90}{1000}} = \frac{50}{27} \text{ molal}$$

dekho kitna asaan hai yr, kyu ghabra rahe ho

153. Concentration of oxygen in ppm can be calculated as:

Concentration of oxygen (ppm)

$$= \frac{\text{mass of oxygen}}{\text{mass of sample}} \times 10^6$$

$$= \frac{3 \times 10^{-3} g}{1000 g} \times 10^6 = 3 \text{ ppm}$$

sawaal nikalo, balle balle karo

158. strength (deals in volume) $\underset{\text{density}}{\overset{\text{density}}{\rightleftarrows}}$ %w / w (deals in weight)

110 gm urea in 1 lt solution (1000 ml)

$$\% \text{ w / w} = \frac{\text{wt. of urea in gm}}{\text{wt. of solution in gm}} \times 100$$

$$= \frac{110 \text{ gm}}{1.1 \frac{\text{gm}}{\text{ml}} \times 1000 \text{ ml}} \times 100 = 10\% \text{ w/w}$$

Bas relation pata hona chaiye, aaram se hoga

159. [1]

$$NaI_{(aq)} + AgNO_{3(aq)} \rightarrow AgI_{(s)} + NaNO_3(aq)$$

20 mL (Excess) 4.74 g

number of moles of I^- in NaI = number of moles of (I^-) in $AgI = \frac{4.74}{235}$

number of moles of $NaI = \frac{4.74}{235}$

Molarity $[NaI] = \frac{4.74}{235 \times 0.02} = 1.008$

160. (C)

$$CaCO_3 + 2HCl \rightarrow CaCl_2 + CO_2 + H_2O$$

Moles of HCl = M × V (L) $= 0.76 \times \frac{250}{1000} = 0.19$

Moles of $CaCO_3$ = Mass/molar mass = $\frac{1000}{100} = 10$

Therefore, HCl is the limiting reagent.

According to the reaction, 2 moles of HCl forms = 1 mol of $CaCl_2$

$\therefore$ 0.19 moles of HCl will form = $\frac{0.19}{2}$ moles of $CaCl_2$

Hence, mass of $CaCl_2$ formed

$$= \frac{0.19}{2} \times 111 = 10.545 \text{ g}$$